Homework 5 Report

SAC+BC

1. Training curves (training steps vs. success rate) of your best SAC+BC agents on cube-single and antsoccer-arena.

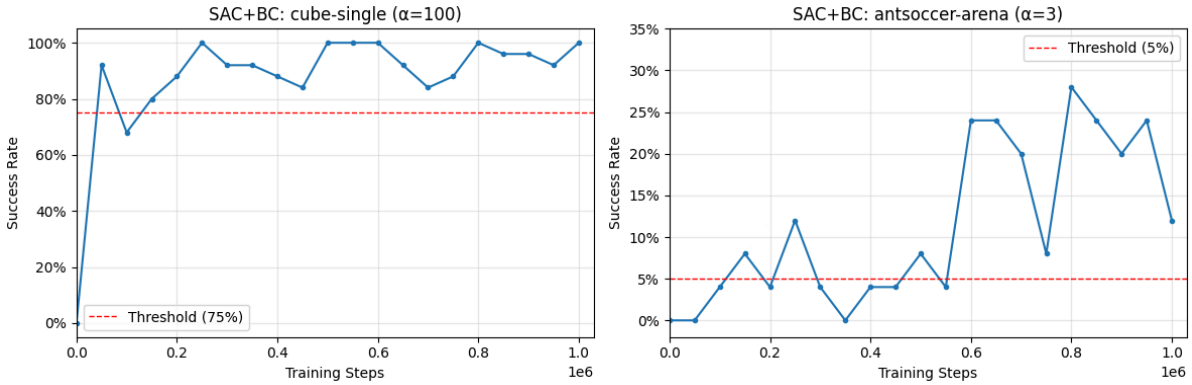


Figure 1: Training curves for the best SAC+BC agents on cube-single-play-singletask-task1-v0 (α=100, left) and antsoccer-arena-navigate-singletask-task1-v0 (α=3, right). The red dashed line indicates the success rate threshold for each task (75% and 5%, respectively). Both agents meet their respective thresholds by 1M training steps.

2. Training curves with different values of α on cube-single

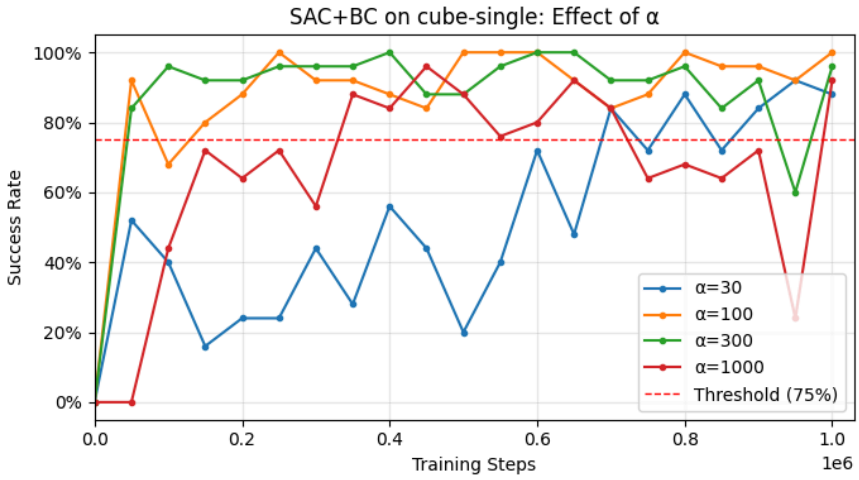


Figure 2: Effect of coefficient α on SAC+BC performance on cube-single-play-singletask-task1-v0. Higher α values means stronger behavioral cloning regularization. All values of α {30, 100, 300, 1000} exceed the 75% threshold at 1M steps.

α values of 30 and 1000 performs worse than α values of 100 and 30, suggesting that insufficient or over regularization leads to suboptimal performance on this task.

IQL

1. Logs of the best-performing IQL agents on the following tasks:
    a. cube-single-play-singletask-task1-v0
    b. antsoccer-arena-navigate-singletask-task1-v0

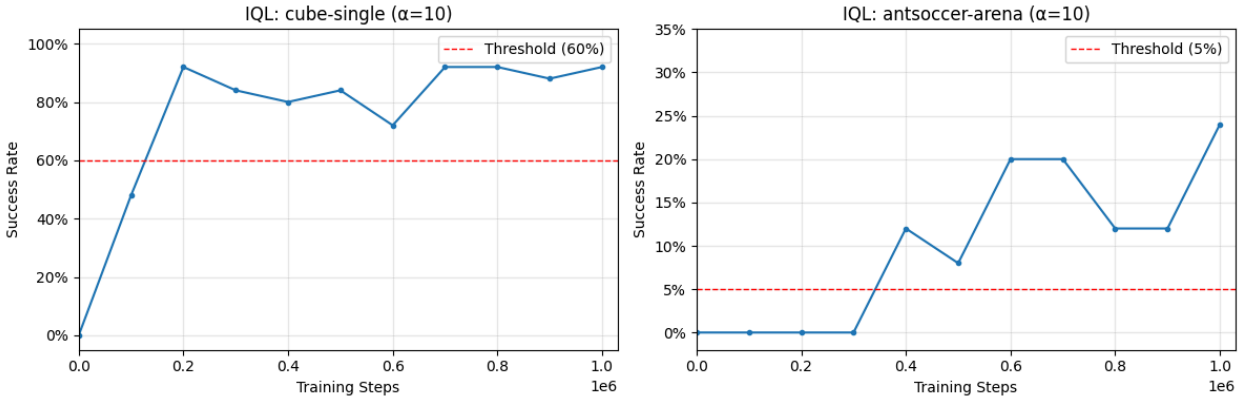


Figure 3: Training curves for the best IQL agents on cube-single-play-singletask-task1-v0 (α=10, left) and antsoccer-arena-navigate-singletask-task1-v0 (α=10, right). The red dashed line shows the success rate threshold for each task (60% and 5%, respectively). Both agents exceed their thresholds by 1M training steps.

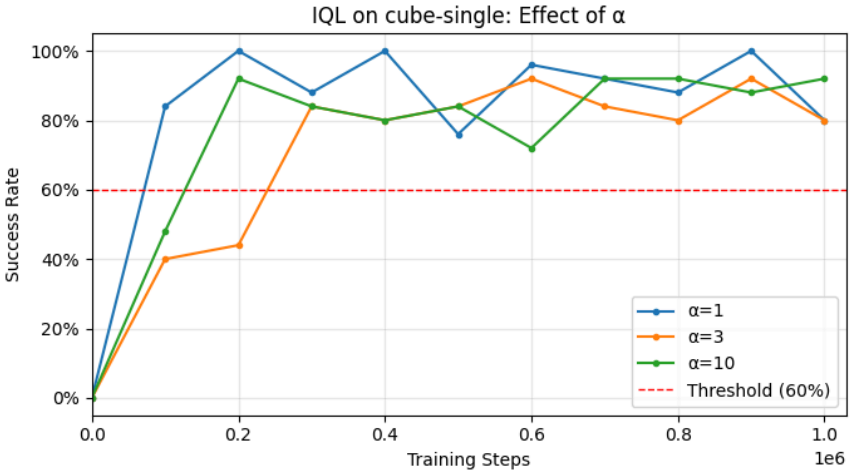


Figure 4: Effect of α on IQL performance on cube-single-play-singletask-task1-v0. All three values α ∈ {1, 3, 10} exceed the 60% threshold, with higher α achieving slightly better final performance (at 1M step).

IQL achieves comparable or better final performance than SAC+BC on both tasks (92% vs. 100% on cube-single, 24% vs. 12% on antsoccer). Sensitivity-wise, even though figure 2 and figure 4 shows that IQL is way less sensitive to α because it converges more stably across training steps, it is incomparable because it sweeps through different α and scales of α.

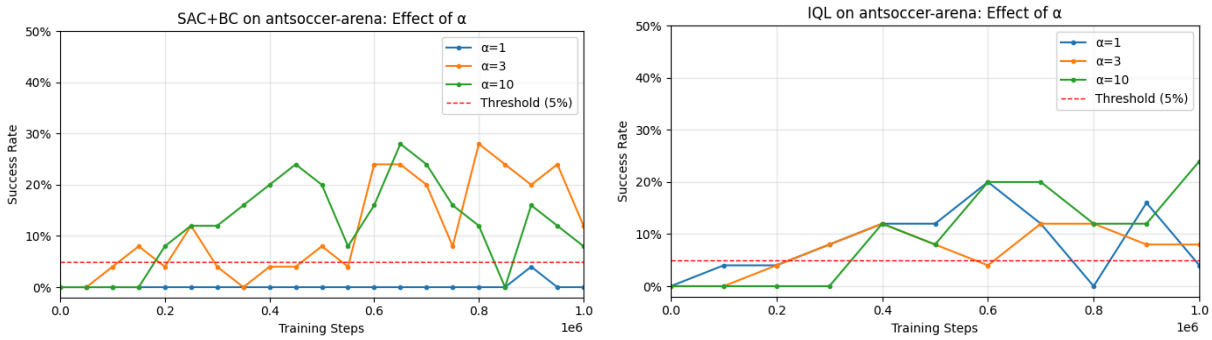


Figure 5: Effect of α on SAC+BC(left) and IQL(right) performance on antsoccer-arena.

We could clearly see that IQL training is more consistent across different α, where SAC+BC is more sensitive to α. SAC+BC's antsoccer performance at different α fluctuates and α = 1 collapses to 0% by the end of training.

FQL

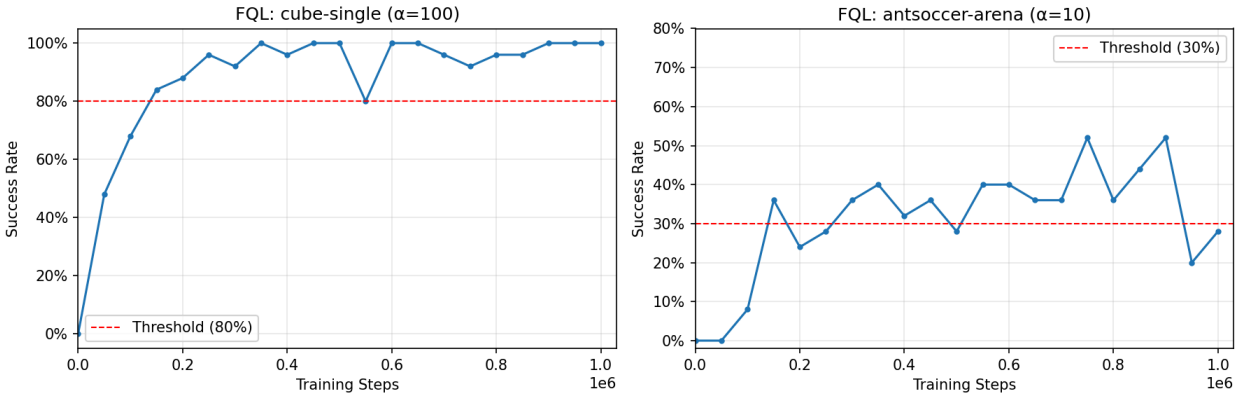


Figure 6: Training curves for the best FQL agents on cube-single-play-singletask-task1-v0 (α=100, left) and antsoccer-arena-navigate-singletask-task1-v0 (α=10, right). The red dashed line shows the success rate threshold for each task (80% and 30%, respectively).

The cube-single agent reaches 100% success rate well before 1M steps. The antsoccer agent peaks at 52% within 1M steps, exceeding the 30% threshold. The eval/success_rate of antsoccer-arena (α=10) reached 0.72 at 1050000 training steps but not included in the figure above.